

Calculus I Lesson 09

1) $f(x) = \frac{x^2 - 1}{x + 1}$

Note: $\frac{x^2 - 1}{x + 1} = \frac{(x + 1)(x - 1)}{(x + 1)} = x - 1$

a) $f(-1) = \underline{\hspace{2cm}}?$

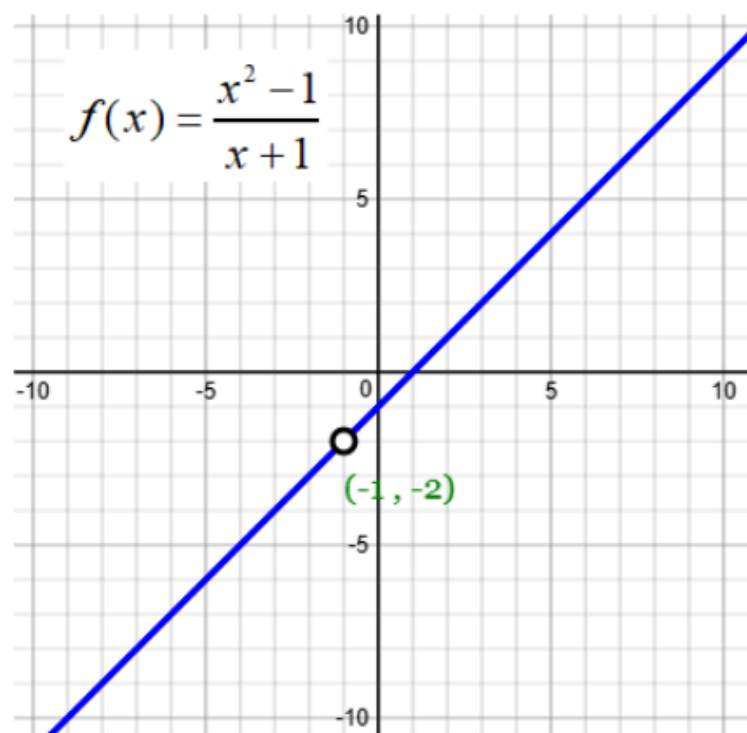
b) $\lim_{x \rightarrow -1^-} f(x) = \underline{\hspace{2cm}}?$

c) $\lim_{x \rightarrow -1^+} f(x) = \underline{\hspace{2cm}}?$

d) $\lim_{x \rightarrow -1} f(x) = \underline{\hspace{2cm}}?$

e) Is $f(-1)$ equal to $\lim_{x \rightarrow -1} f(x)$?

f) Is the function $f(x) = \frac{x^2 - 1}{x + 1}$ continuous at $x = -1$?



$$2) f(x) = \frac{x^2 + 4x - 5}{x - 1}$$

$$\text{Note: } \frac{x^2 + 4x - 5}{x - 1} = \frac{(x + 5)(x - 1)}{(x - 1)} = x + 5$$

$$a) f(1) = \underline{\hspace{2cm}}?$$

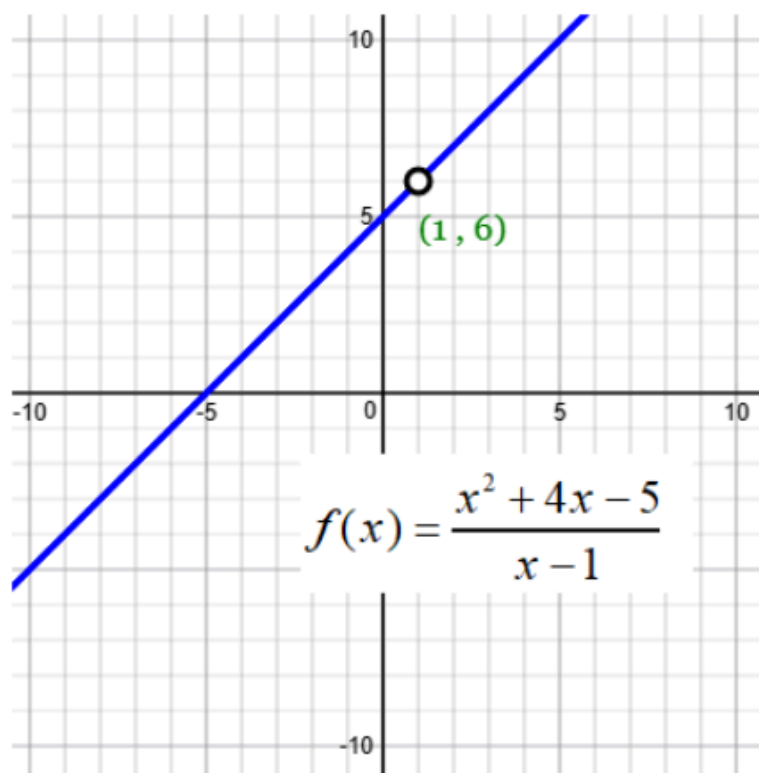
$$b) \lim_{x \rightarrow 1^-} f(x) = \underline{\hspace{2cm}}?$$

$$c) \lim_{x \rightarrow 1^+} f(x) = \underline{\hspace{2cm}}?$$

$$d) \lim_{x \rightarrow 1} f(x) = \underline{\hspace{2cm}}?$$

$$e) \text{ Is } f(1) \text{ equal to } \lim_{x \rightarrow 1} f(x)?$$

$$f) \text{ Is the function } f(x) = \frac{x^2 + 4x - 5}{x - 1} \text{ continuous at } x = 1?$$



$$3) f(x) = \frac{2x^2 - 3x - 2}{x - 2}$$

$$\text{Note: } \frac{2x^2 - 3x - 2}{x - 2} = \frac{(2x + 1)(x - 2)}{(x - 2)} = 2x + 1$$

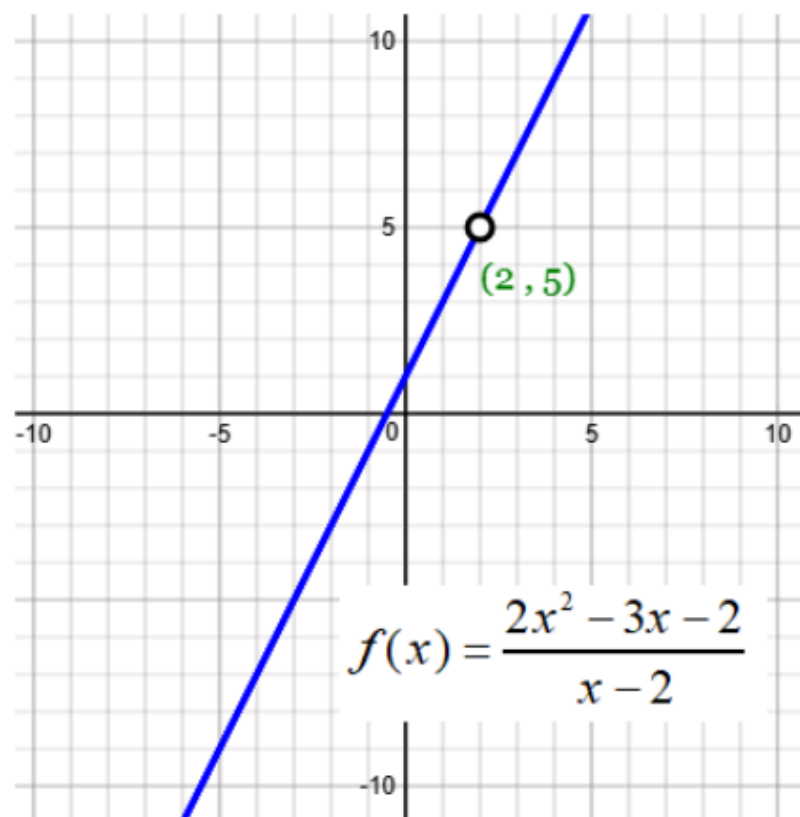
$$a) f(2) = \underline{\quad? \quad}$$

$$b) \lim_{x \rightarrow 2^-} f(x) = \underline{\quad? \quad} \quad c) \lim_{x \rightarrow 2^+} f(x) = \underline{\quad? \quad}$$

$$d) \lim_{x \rightarrow 2} f(x) = \underline{\quad? \quad}$$

$$e) \text{ Is } f(2) \text{ equal to } \lim_{x \rightarrow 2} f(x)?$$

$$f) \text{ Is the function } f(x) = \frac{2x^2 - 3x - 2}{x - 2} \text{ continuous at } x = 2?$$



$$4) f(x) = \frac{2x^2 - 5x - 12}{x - 4}$$

$$\text{Note: } \frac{2x^2 - 5x - 12}{x - 4} = \frac{(2x + 3)(x - 4)}{(x - 4)} = 2x + 3$$

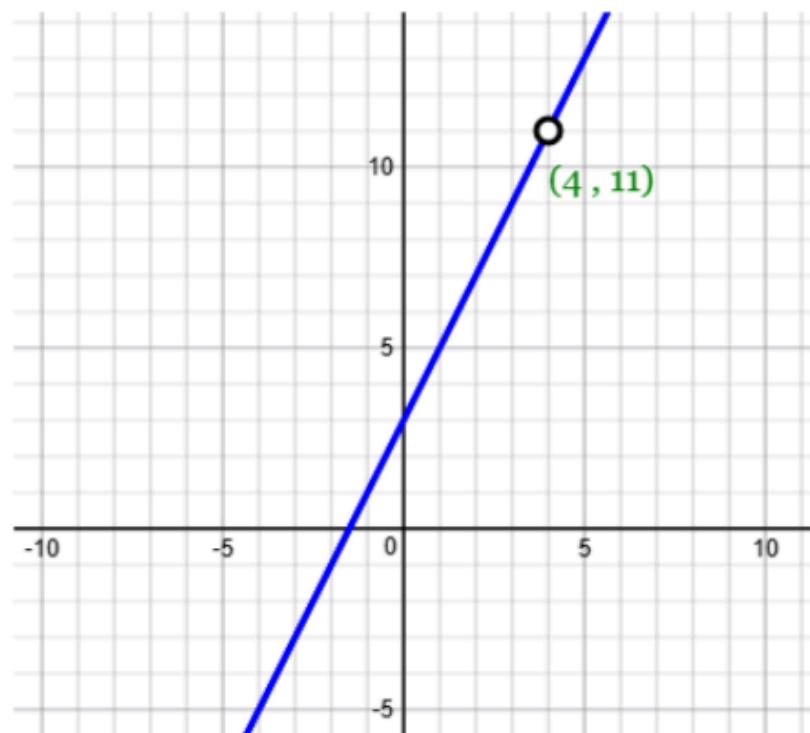
$$a) f(4) = \underline{\hspace{2cm}}?$$

$$b) \lim_{x \rightarrow 4^-} f(x) = \underline{\hspace{2cm}}? \quad c) \lim_{x \rightarrow 4^+} f(x) = \underline{\hspace{2cm}}?$$

$$d) \lim_{x \rightarrow 4} f(x) = \underline{\hspace{2cm}}?$$

$$e) \text{ Is } f(4) \text{ equal to } \lim_{x \rightarrow 4} f(x)?$$

$$f) \text{ Is the function } f(x) = \frac{2x^2 - 5x - 12}{x - 4} \text{ continuous at } x = 4?$$



$$5) f(x) = \begin{cases} 3 - x & \text{if } x \neq 2 \\ 4 & \text{if } x = 2 \end{cases}$$

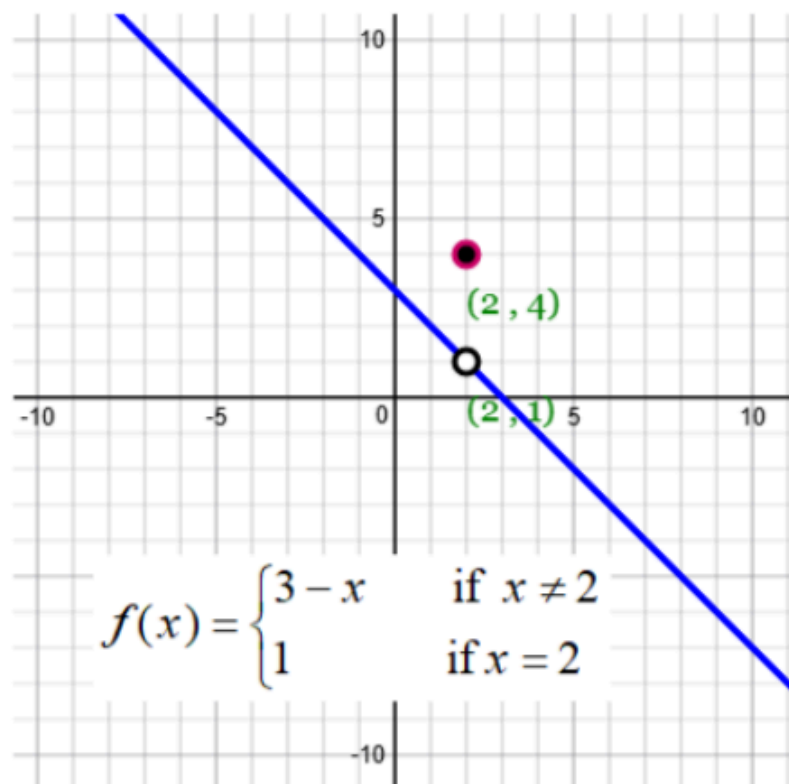
a) $f(2) = \underline{\hspace{2cm}} ?$

b) $\lim_{x \rightarrow 2^-} f(x) = \underline{\hspace{2cm}} ?$ c) $\lim_{x \rightarrow 2^+} f(x) = \underline{\hspace{2cm}} ?$

d) $\lim_{x \rightarrow 2} f(x) = \underline{\hspace{2cm}} ?$

e) Is $f(2)$ equal to $\lim_{x \rightarrow 2} f(x)$?

f) Is the function $f(x) = \begin{cases} 3 - x & \text{if } x \neq 2 \\ 4 & \text{if } x = 2 \end{cases}$ continuous at $x = 2$?



$$6) f(x) = \begin{cases} 2 - x^2 & \text{if } x \neq 1 \\ 3 & \text{if } x = 1 \end{cases}$$

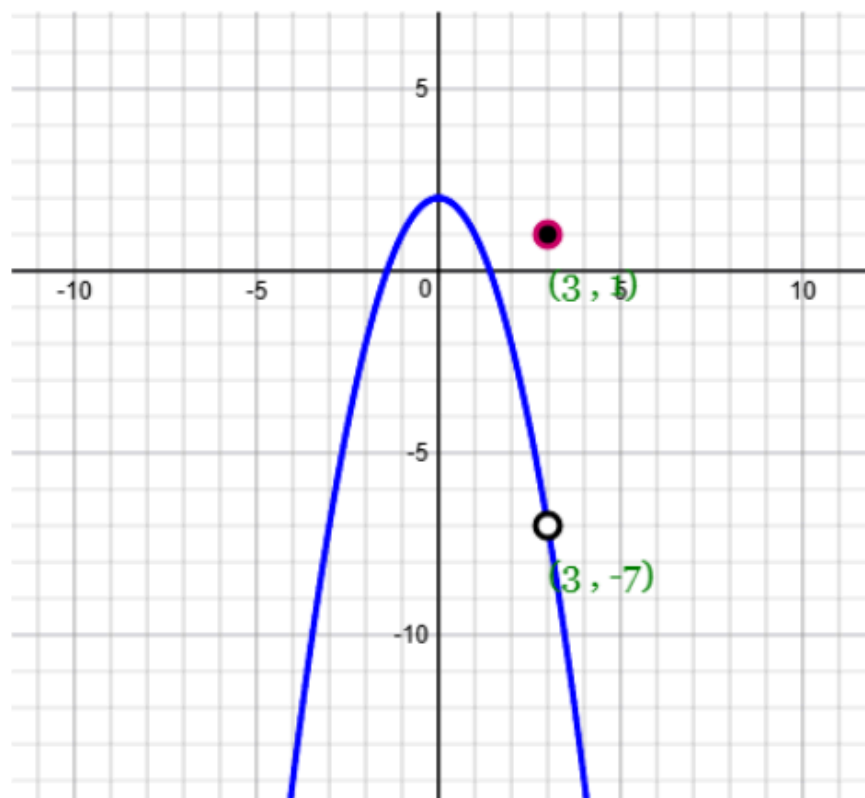
a) $f(1) = \underline{\hspace{2cm}} ?$

b) $\lim_{x \rightarrow 1^-} f(x) = \underline{\hspace{2cm}} ?$ c) $\lim_{x \rightarrow 1^+} f(x) = \underline{\hspace{2cm}} ?$

d) $\lim_{x \rightarrow 1} f(x) = \underline{\hspace{2cm}} ?$

e) Is $f(1)$ equal to $\lim_{x \rightarrow 1} f(x)$?

f) Is the function $f(x) = \begin{cases} 2 - x^2 & \text{if } x \neq 1 \\ 3 & \text{if } x = 1 \end{cases}$ continuous at $x = 1$?



$$7) f(x) = \begin{cases} 2x + 2 & \text{if } x \leq 1 \\ x^2 & \text{if } x > 1 \end{cases}$$

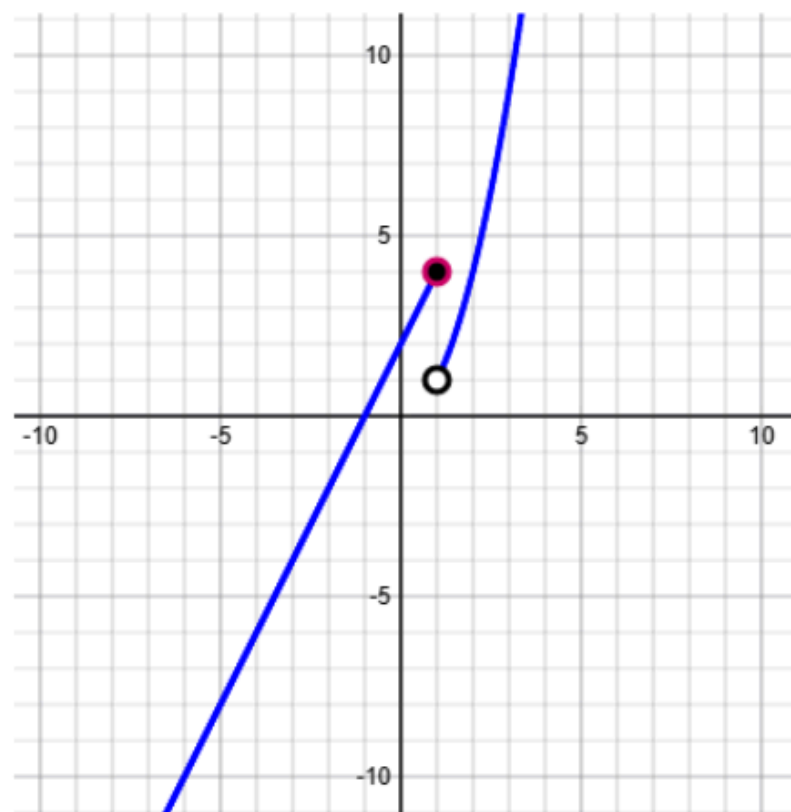
a) $f(1) = \underline{\hspace{2cm}} ?$

b) $\lim_{x \rightarrow 1^-} f(x) = \underline{\hspace{2cm}} ?$ c) $\lim_{x \rightarrow 1^+} f(x) = \underline{\hspace{2cm}} ?$

d) $\lim_{x \rightarrow 1} f(x) = \underline{\hspace{2cm}} ?$

e) Is $f(1)$ equal to $\lim_{x \rightarrow 1} f(x)$?

f) Is the function $f(x) = \begin{cases} 2x + 2 & \text{if } x \leq 1 \\ x^2 & \text{if } x > 1 \end{cases}$ continuous at $x = 1$?



8) $f(x) = \frac{|x|}{x}$

a) $f(0) = \underline{\hspace{2cm}}?$

b) $\lim_{x \rightarrow 0^-} f(x) = \underline{\hspace{2cm}}?$ c) $\lim_{x \rightarrow 0^+} f(x) = \underline{\hspace{2cm}}?$

d) $\lim_{x \rightarrow 0} f(x) = \underline{\hspace{2cm}}?$

e) Is $f(0)$ equal to $\lim_{x \rightarrow 0} f(x)$?

f) Is the function $f(x) = \frac{|x|}{x}$ continuous at $x = 0$?

